

GENE REGULATORY NETWORK INFERENCE FROM INTEGRATIVE MULTI-OMICS DATA

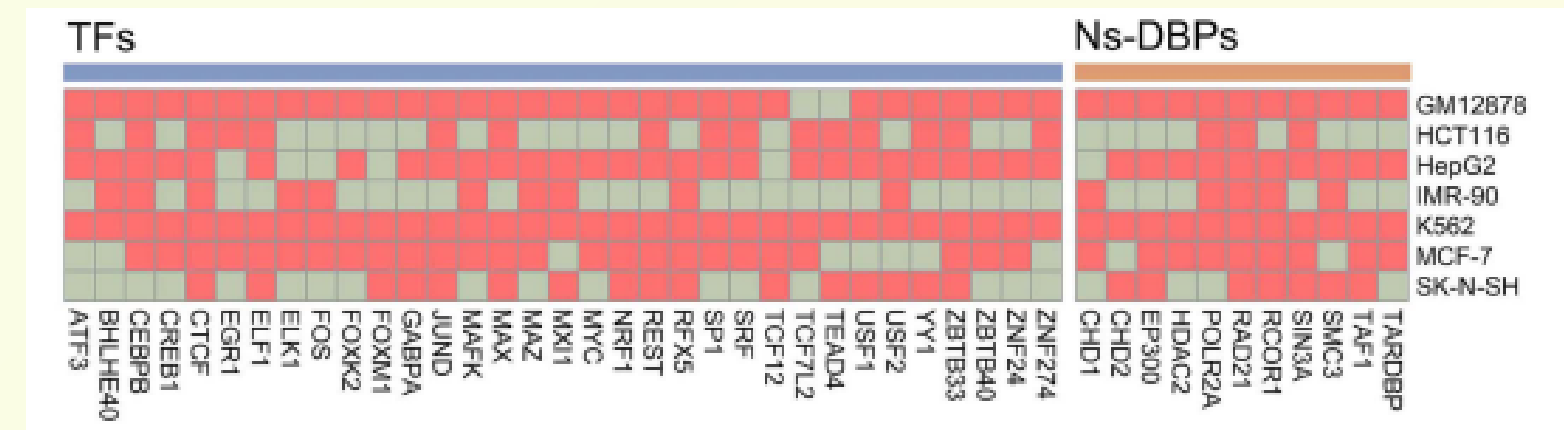
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Paper 1 - EpiXFormer

A Cross-Attention Neural Network for Predicting Cell Type-Specific
Transcription Factor Binding Sites

Datasets



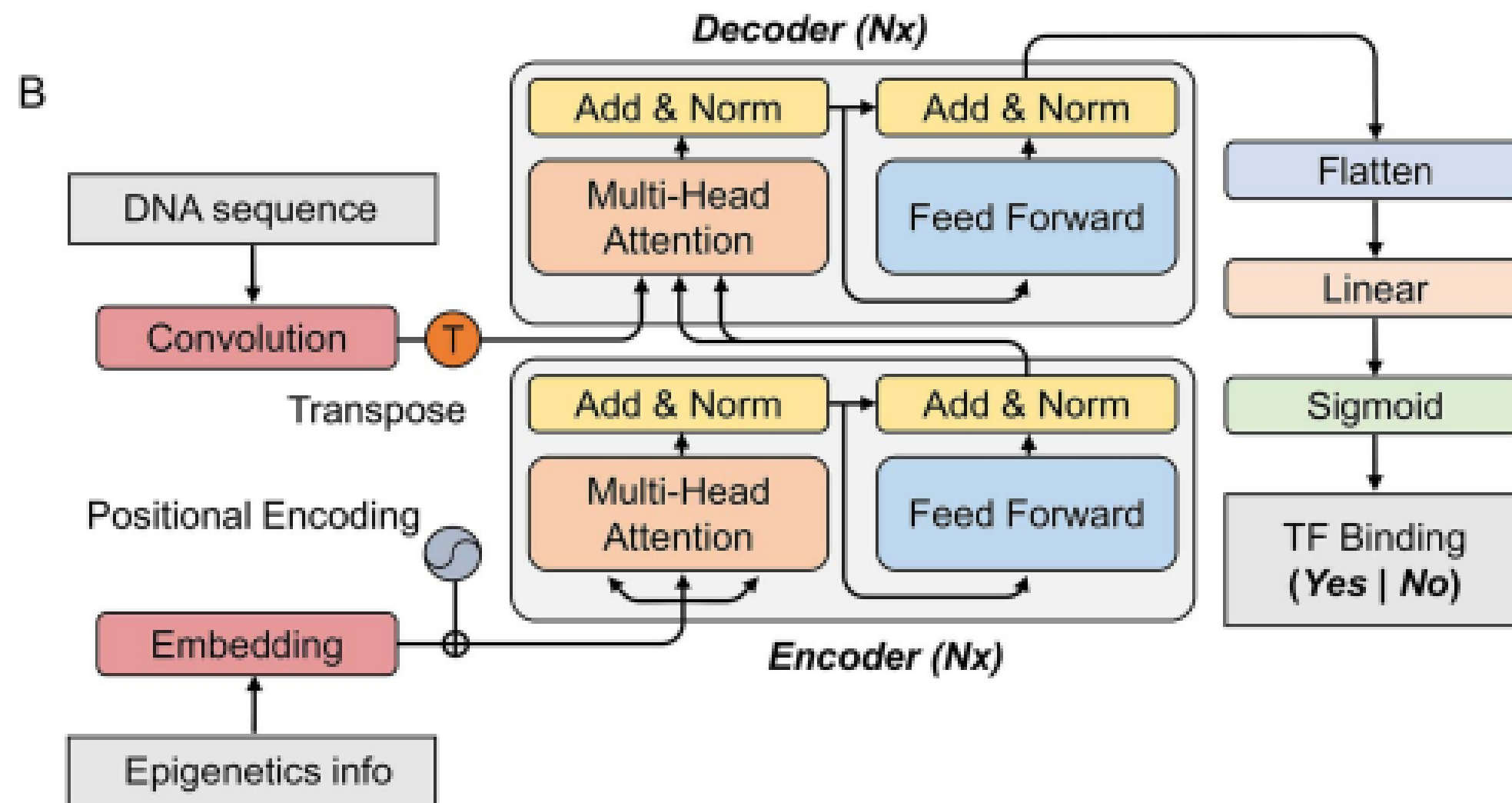
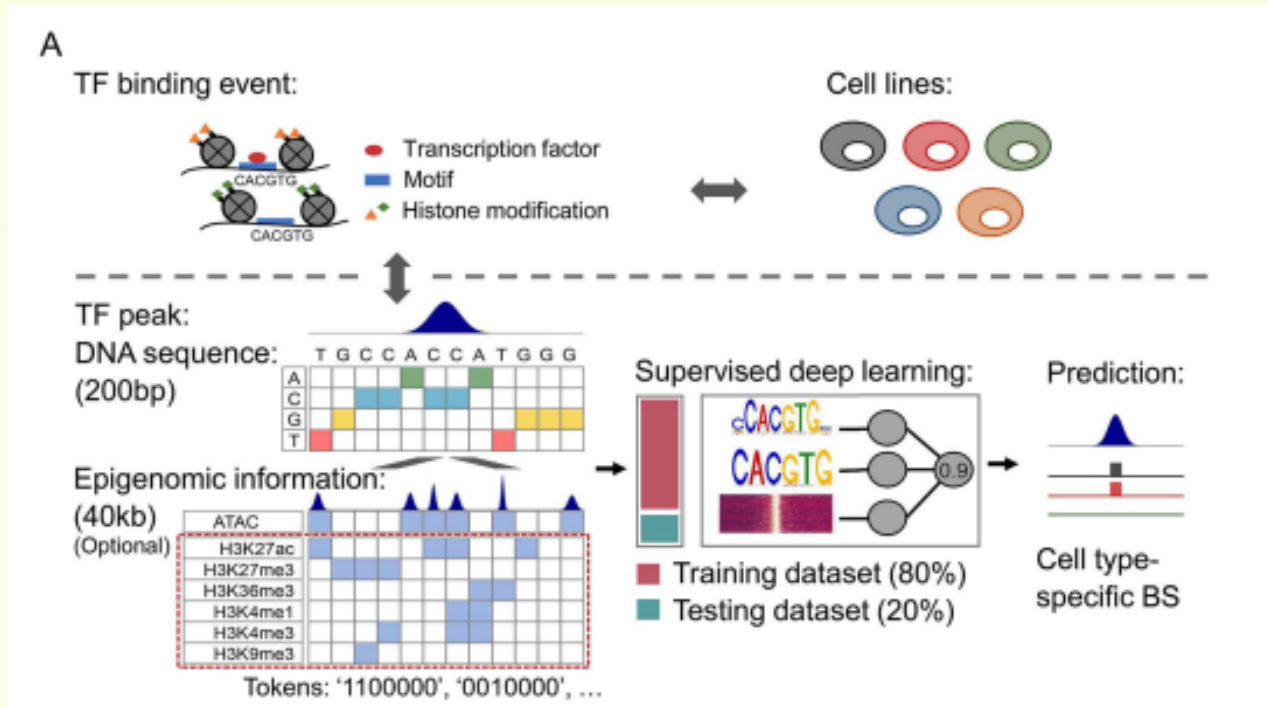
- **Data Source:** Models were trained on publicly available single-cell and bulk multi-omics data collected from the ENCODE portal.
- **Scope:** Analyzed 199 combinations of 43 DNA-binding proteins (DBPs) across 7 distinct cell lines (e.g., GM12878, K562, MCF-7, HepG2).
- **Input Features:** Data consisted of 200bp DNA sequences paired with seven types of epigenomic information, including chromatin accessibility (ATAC-seq) and six histone modifications (e.g., H3K27ac, H3K4me3).
- **Ground Truth:** Positive binding sites were defined by ChIP-seq peaks, while negative sites were generated using FIMO motif scanning or randomly sampled genomic regions lacking DBP binding.

Training & Testing Methodology

- **Task Formulation:** TFBS prediction was framed as a supervised deep learning binary classification problem (predicting TF Binding as Yes or No).
- **Data Split:** Samples for each DBP-cell pair were pooled and randomly partitioned into an 80% training/validation set and a 20% test set.
- **Evaluation Metrics:** Model performance was systematically measured using AUROC, AUPRC, Accuracy, Recall, Precision, F1 score, and Matthews Correlation Coefficient (MCC).
- **Model Variations:** Evaluated base models (all features), semi-models (partial epigenomic data like ATAC-only), and cross-cell models to assess broad generalization capability.

Model Architecture

- **Core Architecture:** EpiXFormer utilizes an encoder-decoder framework.
- **Sequence Processing:** A Convolutional Neural Network (CNN) module processes the primary one-hot encoded DNA sequence inputs.
- **Epigenomic Encoding:** The encoder uses multi-head self-attention and positional encoding to capture the surrounding 40kb context of epigenomic states.
- **Cross-Attention Integration:** The decoder aligns the two modalities via a cross-attention mechanism, where CNN sequence outputs act as queries to attend to the epigenomic context.
- **Output Layer:** The multiscale embedding is fed into a fully connected linear layer with a sigmoid activation function to output final binding probabilities.



Benchmarks & Results

- **High Accuracy:** EpiXFormer base-models achieved a mean AUROC of 0.9905 for sequence-specific TFs and 0.9904 for non-sequence-specific DBPs.
- **Superiority Over Baselines:** Consistently outperformed state-of-the-art sequence-based and multi-modal methods, including DeepBind, FIMO, FactorNet, DeepGRN, Leopard, and maxATAC.
- **Cell-Type Specificity:** Successfully recapitulated the highly distinct, cell type-specific binding landscapes for critical architectural proteins like CTCF and EP300 across different cell lines.
- **Feature Importance:** Semi-model testing revealed that while DNA sequence drives primary binding preference, chromatin status (especially ATAC-seq data) is crucial for defining cell type-specific binding events.

Ablation Studies

Model Input Features	CTCF AUROC x 10	CTCF AUPRC x 10	EP300 AUROC x 10	EP300 AUPRC x 10
Base Model (DNA + 7 Epigenomic marks)	9.942	9.872	9.938	9.867
ATAC Model (DNA + ATAC-seq)	9.936	9.863	9.934	9.822
Epi-core Model (DNA + 5 Histone marks)	9.888	9.767	9.709	9.532
H3K27ac Model (DNA + H3K27ac)	9.861	9.702	9.614	9.378
DNA-only Model (Only DNA sequence)	9.858	9.718	9.047	8.212

Paper 2 - SCMGOT

Deciphering Hierarchical Regulatory Network of Cell Fate via an
Epigenetics-Informed Heterogeneous Graph Transformer

Datasets & Ground Truth

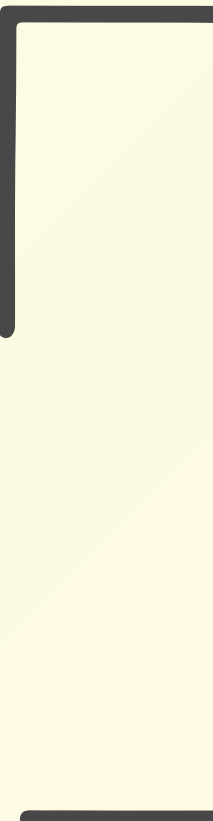
- **Primary Data:** Utilized paired single-cell RNA-seq (scRNA-seq) and single-cell ATAC-seq (scATAC-seq) datasets spanning Bone Marrow, PBMC, human cerebral cortex, leukemia stem cells, and diverse cell lines (K562, HCT116, A549, GM12878).
- **Prior Knowledge Integration:** Incorporated seven extensive prior network databases, bridging epigenomics (ENCODE, JASPAR) and protein-protein interactions (STRING, BioGRID).
- **Ground Truth:** Validated network construction accuracy against rigorous experimental benchmarks, specifically ChIP-seq for TF-CRE binding and Hi-C seq for long-range chromatin conformation (CRE-CRE).
- **Clinical Data:** Leveraged public GWAS and eQTL data (Blood eQTL, GTEx) to link identified regulatory modules to human disease pathology

Training & Testing Methodology

- **Semi-Supervised Learning Strategy:** Addressed the sparsity of known network edges by dynamically sampling highly correlated TF-CRE and CRE-CRE pairs as pseudo-positive labels, alongside controlled pseudo-negative edges.
- **Dual-Task Optimization:** The model learns node representations and network topology simultaneously using a combined loss function.
- **Loss Function Setup:** Utilized Mean Squared Error (MSE) to optimize gene expression predictions and Balanced Binary Cross-Entropy (BLCE) to manage class imbalance during edge classification.
- **Evaluation Metrics:** Assessed edge prediction accuracy using AUPR and AUC against biological ground truths, and utilized Pearson Correlation Coefficient (PCC) and GWAS/eQTL enrichment scores to evaluate downstream predictive tasks.

Model Architecture

- **The SMOGT Framework:** Single-cell Multi-Omics Graph Transformer (SMOGT) utilizes a Hierarchical Epigenetic Graph Transformer as its core.
- **Epigenetic Meta-Path:** Structures information flow strictly along a biological cascade: $\text{TF-TF} \rightarrow \text{TF-CRE} \rightarrow \text{CRE-CRE} \rightarrow \text{CRE-TG}$.
- **Heterogeneous Encoders:** Uses Graph Convolutional Networks (GCN) for modeling TF-TF synergy, Heterogeneous Graph Transformer (HGT) for TF-CRE/CRE-CRE/CRE-TG layers, and GraphSAGE for internal node loops.
- **Downstream Modules:**
 - BioStreamNet: Predicts target gene expression and performs in silico perturbations.
 - MRWR_Pert & MRWR_CTS: Applies multi-layer random walks to track perturbation impacts and identify driver regulators.
 - Louvain_Co-CRE: Discovers clustered cis-regulatory subnetworks.

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Benchmarks & Results

- **Network Accuracy:** SMOGT demonstrated significantly higher accuracy (AUPR and F1 scores) in predicting ChIP-seq validated TF-CRE regulatory bindings compared to SOTA algorithms like regX, LINGER, and SCENIC+.
- **Chromatin Conformation:** Outperformed sliding-window-based algorithms (Cicero and ArchR) in predicting long-range CRE-CRE interactions validated by Hi-C data.
- **Gene Expression Prediction:** The BioStreamNet module achieved the highest Pearson correlation for predicting target gene expression across Bone Marrow, PBMC, and A549 datasets, proving the superiority of its mechanistically informed scaffold.
- **Driver Identification:** Accurately reconstructed lineage-specific networks for Hematopoietic Stem Cell (HSC) branching, successfully identifying literature-validated driver TFs (e.g., KLF1 in the erythroid lineage) where other models failed.



THANK YOU!

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